

This exam paper is part of a set used to assess student abilities as follows:

		Question
CLO 1	Enable to find a truth value of the complex propositions using the truth table. Verify if a complex proposition is tautology.	Q1-Q2
CLO 2	Enable to classify whether an argument is valid or not.	Q3
CLO 3	Enable to convert base numbers to the desired bases and algebraically operate them.	Q4-Q6
CLO 4	Enable to solve absolute inequalities and nonlinear algebraic equations	Q7-Q8
CLO 5	Have moral and ethical values, respect rights, listen to others' opinions, and able to work collaboratively with others.	-

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Examiner

Question 1.

1.1 Suppose that $[(p \rightarrow q) \wedge (p \vee r)] \rightarrow (s \rightarrow r)$ is false.

Find the truth value of $[(p \leftrightarrow q) \wedge (q \leftrightarrow r)] \vee (r \leftrightarrow s)$.

(6 points)

1.2 Determine whether $(p \rightarrow q) \rightarrow [(p \vee q) \rightarrow r]$ is a tautology or not.

(5 points)

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Question 2.

2.1 Determine whether $(p \rightarrow r) \wedge (q \rightarrow r)$ and $\sim(p \vee q) \vee r$ are logically equivalent or not.

(4 points)

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2.2 Given the universal set $U = \{0, 1, 2\}$.

Determine the truth value of $\forall x[(x > 0) \rightarrow (x^2 + 1 > 2x)]$.

(5 points)

Question 3. Given the argument is in the form

- Hypothesis 1. $p \wedge q$
 2. $\sim p \vee r$
 3. $\sim r$

Conclusion q .

Show whether this argument is valid (reasonable) or not.

(5 points)

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Question 4.

4.1 Convert the number $1AC2_{13}$ to a number in base-10.

(4 points)

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4.2 Convert the number 498.21875_{10} to a number in base-8.

(5 points)

Question 5.

5.1 Evaluate $23110_4 + 10110_2 - 1112_3$ and give your answer in base-5. (6 points)

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5.2 Convert the number 100111010001.011010111_2 to a number in base-16.

(5 points)

Question 6.

6.1 Find the value of $67_8 \times 5.4_8$ in base-8.

(5 points)

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6.2 Find the value of $1430_5 \div 31_5$ in base-5.

(5 points)

Question 7. Find all solutions of the absolute value inequality $\left| \frac{x}{x+1} - 1 \right| \leq \frac{1}{2}$.

(9 points)

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Question 8.

8.1 Solve the following nonlinear equation

$$(x^2 - 3x) + \frac{40}{(x^2 - 3x)} = 14.$$

(6 points)

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8.2 Find the sum of all solutions of the equation $10x^3 - 9x^2 - 13x + 6 = 0$. (5 points)